

KWONSEUNG CHO

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EDUCATION

Gwangju Institute of Science and Technology, Gwangju, South Korea

Sep.2023 - present

Ph.D. in Mechanical Engineering

- Fields of Specialty: Controls & Systems, Robotics, Trajectory Optimization, HRI

- Advisor: Dr. Pilwon Hur

Gwangju Institute of Science and Technology, Gwangju, South Korea

Mar.2021 - Aug.2023

M.S. in Mechanical Engineering

- Fields of Specialty: Controls & Systems, Robotics, Trajectory Optimization

- Advisor: Dr. Pilwon Hur

Pukyong National University, Busan, South Korea

Mar.2013 - Aug.2020

B.S. in Control and Instrumentation Engineering

- Fields of Specialty: Controls & Systems, Circuit Theory, Signal Processing

- Advisor: Dr. Gi-Sig Byeon

RESEARCH EXPERIENCE

Gwangju Institute of Science and Technology

Gwangju, South Korea

Research Assistant

Jan2021–Present

Developed a unified control framework integrating action, estimation, and learning via the Free Energy Principle (FEP).

Designed and controlled a lower-limb exoskeleton for hemiparetic stroke patients.

Analyzed and optimized toe-joint mechanics to achieve human-like walking in robots.

Modeled trust in physically interacted human-robot systems to improve HRI quality.

PUBLICATIONS AND PRESENTATIONS

Journal Paper (Under Review and In Prep)

- a. **Kwonseung Cho**, Kangwoo Lee, and Pilwon Hur, “Optimizing Toe Joint Stiffness to Improve Human-Like Walking,” *Scientific Reports*, — under review

Journal Paper (Published)

- a. Veronica Knisley, **Kwonseung Cho**, Kang-Woo Lee, and Pilwon Hur, Preferred Jacobian Differentiation and Direct Collocation Methods for an Efficient and Accurate Walker Gait Optimization, *International Journal of Control, Automation, and Systems*, Vol 23, Issue 5, pp1551-1562, 2025

Conference Papers and Presentations

- a. **Kwonseung Cho**, Jiyeon Sung, Kangwoo Lee, Sunghyun Kim, and Pilwon Hur, “Development of a Lower Limb Exoskeleton for Gait Rehabilitation in Hemiparetic Patients”, *40th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Jeonju, South Korea, Jun25–7,2025
- b. **Kwonseung Cho**, Kang-Woo Lee, and Pilwon Hur, “Optimizing Toe Joint Stiffness to Improve Human-Like Walking,” *Spring Conference of the Bio Division, Korean Society of Mechanical Engineers (KSME)*, Busan, South Korea, May, 2025
- c. Sunwoong Moon, Myeongju Cha, Hyungseok Ryu, **Kwonseung Cho**, Jiyeon Sung, Kangwoo Lee, Sunghyun Kim, and Pilwon Hur, “State Estimation of Nonlinear Systems Based on the Free Energy Principle,” *39th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Daejeon, South Korea, July2–4,2024
- d. **Kwonseung Cho**, and Pilwon Hur, “Effect of Toe Joint on Bipedal Walking”, *2023 Conference of the Korean Society of Biomechanics*, Jeonju, South Korea, 2023

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- e. Veronica Knisley, **Kwonseung Cho**, and Pilwon Hur, “Preferred Jacobian Differentiation and Direct Collocation Method for an Efficient and Accurate Bipedal Walking Gait,” *38th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Samcheok, South Korea, 2023
 - f. **K. Cho** and P. Hur, “Effect of Toe Joint on Bipedal Walking: A Simulation Study”, *2022 Conference of the Korean Society of Biomechanics*, Chungju, South Korea, 2022
 - g. **K. Cho**, Shawanee Patrick, Namita Anil Kumar, Woo-rim Hong, and Pilwon Hur, “Biomechanical Effect of Toe Joint on Transfemoral Prosthetic Walking,” *Spring Conference of the IT Convergence Division, Korean Society of Mechanical Engineers (KSME)*, Andong, South Korea, May25–27,2022
 - h. **K. Cho**, C. Debuys, and P. Hur, “Minimizing Angular Momentum Yields More Robust Walking Trajectories in a Compass-Gait Walker,” *36th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Yeosu, South Korea, 2021
 - i. S. Moon, M. Cha, **K. Cho**, and P. Hur, “Finding the Optimal Trajectory Using a Direct Collocation Method for an Efficient and Accurate Compass-Gait Walker for Undergraduates,” *36th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Yeosu, South Korea, 2021
 - j. M. Cha, S. Moon, **K. Cho**, and P. Hur, “Dynamics and Control Methods for a Compass-Gait Walker: Tutorial of Defining EOM and Simulation Method for Undergraduates,” *36th Annual Conference of the Institute of Control, Robotics and Systems (ICROS)*, Yeosu, South Korea, 2021
 - k. **K. Cho**, H. Hong, G. Kim, and H. Yun, “Trends and Review of Autonomous Driving Control for Agricultural Robots,” *2020 Fall Conference of the Korean Society of Manufacturing Process Engineers*, Yeosu, South Korea, 2020

AWARDS & HONORS

Best Paper Award, KSAB 2023

Korean government scholarship, Gwangju Institute of Science and Technology(GIST) Spring.2021 - Fall.2022

Leadership Award, Pukyong National University Fall.2018

Company-sponsored Scholarship, Pukyong National University Spring.2018

Company-sponsored Scholarship, Pukyong National University Spring.2016 - Fall.2016

Academic Scholarship, Pukyong National University Spring.2016

Academic Scholarship, Pukyong National University Spring.2013

TEACHING EXPERIENCE

Gwangju Institute of Science and Technology

Gwangju, South Korea

Teaching Assistant

MC 3215 Mechanical Signals and Systems

SP2021, SP2024, SP2025

ME 5160 Advanced Dynamics

Spring 2023

Motor Boot Camp

2024, 2025

MC 3107-01,02 Mechanical Engineering Experiment 2

Fall 2024

Suwan & Jeonnam High Schools – GIST Lab Exploration Program

2025

Gwangju Science High School – Creative Research Program

2025

WORK EXPERIENCES

Agriculture Robotics & Automation Research Center, Korea Institute of Robotics & Technology Convergence (KIRO)

Andong, South Korea

Researcher

Sep.2020 - Feb.2021

- Modeling of High DOF Mobile Robot (Farm-bot)

- Configuring Gazebo simulation environment

- collecting and analysis sensor data using ROS

Smart Mobility Research Lab at Electronics and Telecommunication Research Institute (ETRI)

Research Assistance

July.2020 - Aug.2020

- Torque Control for 6DOF Robot Manipulators / Kinematics and Dynamics Calculation
- Configuring the simulation environment (ROS, Gazebo, Moveit)
- Teaching Robot Manipulator with RL

Transportation Research Lab at University of Nevada, Las Vegas

Research Assistance

Aug.2019 - Dec.2019

- Machine Learning in Micro-controller
- ROS based indoor mobile robot design
- IoT system design for Smart Home System

User System Department, Kyushu Association of Independent Entrepreneurs at Fukuoka, Japan

Intern

Jan.2019 - Feb.2019

- User data collecting analysis

TECHNICAL STRENGTHS

Operating System	Windows, Mac, Linux, ROS (Robot Operating System)
Computer Language	C++, Python, Julia
Engineering Tool	MATLAB, Mathematica, LaTeX
Embedded System	Arduino, Raspberry Pi, Jetson, BeagleBone Black

PROFESSIONAL SERVICE ACTIVITIES

Student Organization

Control and Instrumentation Engineering Student Organization at Pukyong National University

- President

Spring.2018 - Fall.2018

EXTRA-CURRICULAR ACTIVITIES

Study abroad,

Intensive English Institute, University of Illinois, Urbana and Champaign

Sep.2016 - July.2017

Military Service,

Republic of Korea Airforce (ROK Airforce)

July.2013 - July.2015